

# CHARLES BENELLO

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## EDUCATION

### The University of Chicago

Master of Science in Financial Mathematics

- Wayne Booth Graduate Student Teaching Prize (2026) - UChicago's top graduate teaching award
- Coursework: Options Pricing, Probability & Stochastic Processes, Portfolio & Risk Management, Computing for Finance, ML in Finance

Chicago, IL

Expected Dec 2026

### The University of Chicago | GPA: 3.75/4.0

Bachelor of Science in Mathematics and Computer Science (Systems), Minor in Korean

- Coursework: Algorithms, Database Systems, Distributed Systems, Concurrency Control, Operating Systems

Chicago, IL

Jun 2025

## TECHNICAL SKILLS

**Languages:** Python, Rust, C/C++, SQL, CUDA, TypeScript, Go

**Frameworks & Tools:** PyTorch, XGBoost, NumPy/Pandas, scikit-learn, Docker, Git, Linux, PostgreSQL, DuckDB, Databricks

**Concepts:** Options Pricing, Stochastic Processes, Portfolio Optimization, Statistical Modeling, Machine Learning, Distributed Systems, GPU Programming

## EXPERIENCE

### Quantitative Researcher

Mar 2026 – Present

PayPal (UChicago Project Lab)

Chicago, IL

- Built a self-supervised transaction foundation model (TabBERT-style hierarchical transformer, ~5M params) on 14.5M unlabeled transactions; achieved AUC 0.974 / F1 0.822 on fraud detection at 0.12% positive rate, with focus on probability calibration in the tails.
- Ran 87+ ablations on UChicago's Midway3 SLURM cluster; implemented per-field masking and SPRM from PANTHER (NeurIPS 2025) and PRAGMA (Revolut 2026), with per-field masking yielding +1.9% AUC and +16.8% F1 over baseline.

### Software Engineering Intern

Feb 2026 – Present

Alter Domus

Chicago, IL

- Lifted document extraction accuracy from 50% to 92–97% by architecting a multi-stage pipeline with schema-enforced structured output and cross-reference verification for 100+ page financial filings.
- Built an LLM-as-a-Judge evaluation framework for faithfulness, numerical accuracy, and completeness; cut LLM inference costs by optimizing context handling.

### Quantitative Researcher

Oct – Dec 2025

Bank of America (UChicago Project Lab)

Chicago, IL

- Built an AI-driven platform to rebalance high-quality liquid asset portfolios, generating trade recommendations that minimize risk under capital, liquidity, and net interest income constraints.
- Engineered a multi-agent LLM debate system where competing agents generate macro stress scenarios and produce probability-weighted reallocation recommendations.

### Database Research Assistant

Apr 2023 – Present

ChiData Research Lab, University of Chicago

Chicago, IL

- First-authored VLDB 2026 paper by designing a parallel external merge sort in Rust that completes under tight memory budgets where PostgreSQL, DuckDB, and ClickHouse abort.
- Sole-authored HPTS 2026 paper on a lightweight prediction model for sort configuration on cloud and disaggregated hardware, transferring calibration across machines within 3–5% of optimal.

### GPU Systems Researcher

Jun – Sep 2025

EPFL (DIAS Lab)

Lausanne, Switzerland

- Cut out-of-memory query times by 50–200% by engineering a macroblock-based GPU decompression strategy in CUDA/C++ for large analytical workloads.

## PROJECTS

### Transit Coverage Optimizer | Python | [cmbenello.github.io/covtsp](https://github.com/cmbenello/covtsp)

2026

- Beat the Guinness World Record by 60 minutes (16h45m vs. 17h45m) by building a Covering TSP solver on a time-expanded graph (160K nodes, 1.26M edges) with k-NN and epsilon-greedy search; generalized to NYC and Berlin.

### Relational Database Engine | Rust | [github.com/cmbenello/CrustyDB](https://github.com/cmbenello/CrustyDB)

2023

- Implemented a multi-threaded relational database from scratch: B+ tree indexing, concurrent buffer pool with page-level locking, heap storage, and a SQL query execution engine.

## PUBLICATIONS

[1] R. Otaki\*, C. Benello\*, et al. "CrocSort: Resource-Efficient, Skew-Resilient Parallel External Merge Sort." *VLDB 2026*. (\*Equal contribution)

[2] C. Benello. "Getting the Most Out of External Sorting in Cloud and Disaggregated Environments." *HPTS 2026*.

[3] R. Otaki, J.H. Chang, C. Benello, A. Elmore, G. Graefe. "Resource-Adaptive Query Execution." *CIDR 2025*.

## ADDITIONAL

**Citizenship:** US, UK, Italy | **Languages:** Korean, Italian (Intermediate); French (Basic) | **Interests:** Running (Chicago Marathon '22, '23, '24), Climbing

**Charles Benello**

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April 27, 2026

## **Akuna Capital – Options 201 Course Application**

I'm a Master's student in Financial Mathematics at UChicago, half a mile from your Loop office. I just finished the Options Pricing course this term. What pulled me toward this side of the field wasn't the math itself but the way the math has immediate, measurable consequences: get the Greeks wrong on a vol surface and you lose money in a way that getting a regression coefficient slightly off in a paper never punishes you for.

A snapshot of this year:

At PayPal, through UChicago's Project Lab, I'm a quantitative researcher working on transaction foundation models for fraud detection. I'm comparing BERT-style masked prediction against GPT-style next-token prediction across 24M+ transactions, with ablations on tokenization, time encoding, and context window size. The hard problem is calibrating probability on a 0.12% positive rate, which is structurally similar to thinking about edge in low-probability tail events.

Last fall at Bank of America (also through Project Lab) I built an AI-driven platform to rebalance HQLA portfolios under capital, liquidity, and NII constraints, with a multi-agent LLM debate system that generates macro stress scenarios and produces probability-weighted reallocation recommendations. At Alter Domus, my current SWE internship, I lifted document extraction accuracy on 100+ page financial filings from a 50% baseline to 92–97% by architecting a multi-stage pipeline with cross-reference verification.

On the systems side: my paper CrocSort (VLDB 2026, co-first-author) builds a parallel external merge sort in Rust that completes within memory budgets where PostgreSQL, DuckDB, and ClickHouse abort. I sole-authored a follow-up at HPTS 2026 on sort configuration for cloud and disaggregated hardware. At EPFL last summer I cut out-of-memory query times by 50 to 200% with a CUDA decompression strategy.

What I think sets me apart from a typical applicant is that I'm an engineer-first quant. Akuna's pitch is that competitive market-making is a software problem as much as it is a math problem, and I've spent three years on the software side of that equation, where a slow or wrong answer has measurable cost. I write performant Rust and C++ as comfortably as I write Python, I've built end-to-end ML pipelines, and I've shipped kernel-level CUDA. I also won the Wayne Booth Graduate Student Teaching Prize this year, UChicago's top graduate teaching award, for the ability to communicate hard concepts under time pressure to rooms of students each pulling for a different answer. I'd argue that's the closest thing in academic life to a trading desk.

The honest reason I want Options 201: I want to figure out whether trading is the right next step before committing. I've spent serious time inside research, full-stack engineering, and ML. I have not spent serious time inside a trading firm. A week of lectures, mini-shadows, and quizzes is the cleanest way for me to find out, and the fact that you're headquartered in Chicago makes the practical case obvious.

Thanks for considering me.

Charles Benello